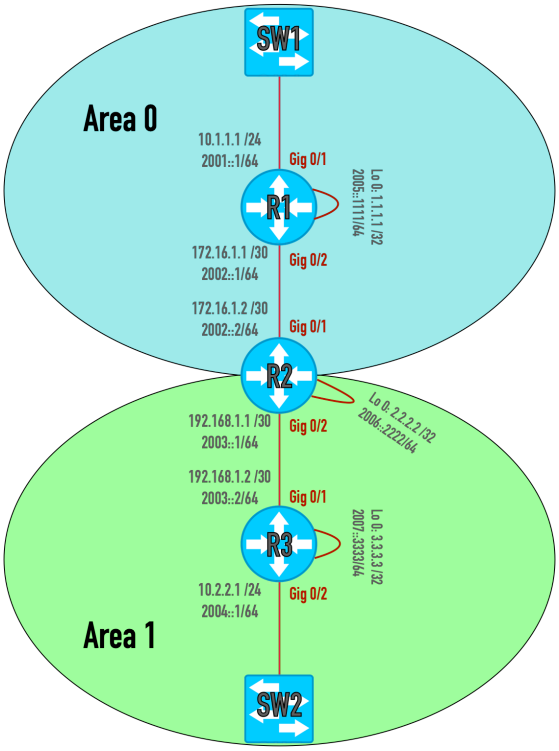
OSPFv3 Address Families Configuration

Topology



Initial Configuration Commands

R1:

```
enable
conf t
host R1
no banner motd
no banner login
no banner exec
no banner incoming
line vty 0 15
password cisco
login
exec-timeout 0 0
line con 0
logging synchronous
no ip-domain lookup
int gig 0/1
ip address 10.1.1.1 255.255.255.0
```

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```
ipv6 address 2001::1/64
no shut
int gig 0/2
ip address 172.16.1.1 255.255.255.252
ipv6 address 2002::1/64
no shut
int lo0
ip address 1.1.1.1 255.255.255.255
ipv6 address 2005::1111/64
end
copy run star
```

R2:

```
enable
conf t
host R2
no banner motd
no banner login
no banner exec
no banner incoming
line vty 0 15
password cisco
login
exec-timeout 0 0
line con 0
logging synchronous
no ip-domain lookup
int lo0
ip address 2.2.2.2 255.255.255.255
ipv6 address 2006::2222/64
int gig 0/1
ip address 172.16.1.2 255.255.255.252
ipv6 address 2002::2/64
no shut
int gig 0/2
ip address 192.168.1.1 255.255.255.252
ipv6 address 2003::1/64
no shut
end
copy run star
```

R3:

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```
enable
conf t
host R3
no banner motd
no banner login
no banner exec
no banner incoming
line vty 0 15
password cisco
login
exec-timeout 0 0
line con 0
logging synchronous
no ip-domain lookup
int lo0
ip address 3.3.3.3 255.255.255.255
ipv6 address 2007::3333/64
int gig 0/1
ip address 192.168.1.2 255.255.255.252
ipv6 address 2003::2/64
no shut
int gig 0/2
ip address 10.2.2.1 255.255.255.0
ipv6 address 2004::1/64
no shut
end
copy run star
```

(##Note, this lab uses the same base configuration and topology as the OSPFv3 Traditional Configuration Lab.)

Lab Tasks

1. Let's enable IPv6 on R1, and configure OSPFv3 using the address families approach. We are going to set up both IPv4 and IPv6.
2. Let's enable IPv6 on R2, and configure OSPFv3 using the address families approach. We are going to set up both IPv4 and IPv6.
3. Let's enable IPv6 on R3, and configure OSPFv3 using the address families approach. We are going to set up both IPv4 and IPv6.
4. On R3, let's see if we can see our neighbor by using the "show ospfv3 neighbor" command.
5. On R3, let's see if we can see the database for each of our address families by using the "show ospfv3 database" command.

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6. On R3, let's see if we can see our IPv4 routes by using the "show ip route" command. Let's also see if we are able to ping R1's IPv4 loopback interface.
7. On R3, let's see if we can see our IPv6 routes by using the "show ipv6 route" command. Let's also see if we are able to ping R1's IPv6 loopback interface.

Solution

Step 1: Let's enable IPv6 on R1, and configure OSPFv3 using the address families approach. We are going to set up both IPv4 and IPv6.

```
R1>en
R1#conf t
R1(config)#ipv6 unicast-routing
R1(config)#ipv6 cef
R1(config)#router ospfv3 1
R1(config-router)#?
Router configuration commands:
  address-family          Enter Address Family command mode
  adjacency               Control adjacency formation
  area                    OSPF area parameters
  authentication          Authentication parameters
  auto-cost               Calculate OSPF interface cost according to bandwidth
  bfd                     BFD configuration commands
  compatible              Compatibility list
  default                 Set a command to its defaults
  event-log               Event Logging
  exit                    Exit from routing protocol configuration mode
  graceful-restart        Graceful-restart options
  help                    Description of the interactive help system
  interface-id            Source of the interface ID
  limit                   Limit a specific OSPF feature
  local-rib-criteria      Enable or disable usage of local RIB as route criteria
  log-adjacency-changes   Log changes in adjacency state
  manet                   Specify MANET OSPF parameters
  max-lsa                 Maximum number of non self-generated LSAs to accept
  max-metric              Set maximum metric
  no                      Negate a command or set its defaults
  passive-interface       Suppress routing updates on an interface
  prefix-suppression      Enable prefix suppression
  queue-depth             Hello/Router process queue depth
  router-id               router-id for this OSPF process
  shutdown                Shutdown the router process
  timers                  Adjust routing timers

R1(config-router)#router-id 1.1.1.1
R1(config-router)#passive-interface gig 0/1
R1(config-router)#passive-interface lo0
R1(config-router)#address-family ipv4
R1(config-router-af)#address-family ipv6
R1(config-router-af)#?
Router Address Family configuration commands:
  adjacency               Control adjacency formation
  area                    OSPF area parameters
  authentication          Authentication parameters
  auto-cost               Calculate OSPF interface cost according to bandwidth
  bfd                     BFD configuration commands
  compatible              Compatibility list
  default                 Set a command to its defaults
```

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```
  default-information    Distribution of default information
  default-metric         Set metric of redistributed routes
  discard-route          Enable or disable discard-route installation
  distance               Administrative distance
  distribute-list        Filter networks in routing updates
  event-log              Event Logging
  exit-address-family    Exit from Address Family configuration mode
  graceful-restart       Graceful-restart options
  help                   Description of the interactive help system
  interface-id           Source of the interface ID
  limit                  Limit a specific OSPF feature
  local-rib-criteria     Enable or disable usage of local RIB as route criteria
  log-adjacency-changes  Log changes in adjacency state
  manet                  Specify MANET OSPF parameters
  max-lsa                Maximum number of non self-generated LSAs to accept
  max-metric             Set maximum metric
  maximum-paths          Forward packets over multiple paths
  no                     Negate a command or set its defaults
  passive-interface      Suppress routing updates on an interface
  prefix-suppression     Enable prefix suppression
  queue-depth            Hello/Router process queue depth
  redistribute           Redistribute IPv6 prefixes from another routing
                         protocol
  router-id              router-id for this OSPF process
  shutdown               Shutdown the router process
  snmp                   Modify snmp parameters
  summary-prefix         Configure IPv6 summary prefix
  table-map              Map external entry attributes into routing table
  timers                 Adjust routing timers

R1(config-router-af)#int gig 0/1
R1(config-if)#ospfv3 1 ipv4 area 0
R1(config-if)#ospfv3 1 ipv6 area 0
R1(config-if)#int gig 0/2
R1(config-if)#ospfv3 1 ipv4 area 0
R1(config-if)#ospfv3 1 ipv6 area 0
R1(config-if)#int lo0
R1(config-if)#ospfv3 1 ipv4 area 0
R1(config-if)#ospfv3 1 ipv6 area 0
R1(config-if)#end
```

(##An address family is going to be a configuration for a specific protocol.)
(##Within address family configuration mode, there are commands that we can give that would apply specifically to the address family we are configuring right now. Under router configuration mode, we can specify a configuration that would be used by both address families.)

Step 2: Let's enable IPv6 on R2, and configure OSPFv3 using the address families approach. We are going to set up both IPv4 and IPv6.

```
R2>en
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#ipv6 unicast-routing
R2(config)#ipv6 cef
```

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```
R2(config)#router ospfv3 1
R2(config-router)#router-id 2.2.2.2
R2(config-router)#passive-interface lo0
R2(config-router)#address-family ipv4
R2(config-router-af)#address-family ipv6
R2(config-router-af)#int gig 0/1
R2(config-if)#ospfv3 1 ipv4 area 0
R2(config-if)#ospfv3 1 ipv6 area 0
R2(config-if)#int gig 0/2
R2(config-if)#ospfv3 1 ipv4 area 1
R2(config-if)#ospfv3 1 ipv6 area 1
R2(config-if)#int lo 0
R2(config-if)#ospfv3 1 ipv4 area 1
R2(config-if)#ospfv3 1 ipv6 area 1
R2(config-if)#end
```

Step 3: Let's enable IPv6 on R3, and configure OSPFv3 using the address families approach. We are going to set up both IPv4 and IPv6.

```
R3>en
R3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#ipv6 unicast-routing
R3(config)#ipv6 cef
R3(config)#router ospfv3 1
R3(config-router)#router-id 3.3.3.3
R3(config-router)#passive-interface lo0
R3(config-router)#passive-interface gig 0/2
R3(config-router)#address-family ipv4
R3(config-router-af)#address-family ipv6
R3(config-router-af)#int gig 0/1
R3(config-if)#ospfv3 1 ipv4 area 1
R3(config-if)#ospfv3 1 ipv6 area 1
R3(config-if)#int gig 0/2
R3(config-if)#ospfv3 1 ipv4 area 1
R3(config-if)#ospfv3 1 ipv6 area 1
R3(config-if)#int lo0
R3(config-if)#ospfv3 1 ipv4 area 1
R3(config-if)#ospfv3 1 ipv6 area 1
R3(config-if)#end
```

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Step 4: On R3, let's see if we can see our neighbor by using the "show ospfv3 neighbor" command.

```
R3#show ospfv3 neighbor

          OSPFv3 1 address-family ipv4 (router-id 3.3.3.3)

Neighbor ID     Pri   State           Dead Time   Interface ID    Interface
2.2.2.2           1   FULL/DR         00:00:31    4               GigabitEthernet0/1

          OSPFv3 1 address-family ipv6 (router-id 3.3.3.3)

Neighbor ID     Pri   State           Dead Time   Interface ID    Interface
2.2.2.2           1   FULL/DR         00:00:32    4               GigabitEthernet0/1
```

(##Notice that we have output for each of our address families.)

Step 5: On R3, let's see if we can see the database for each of our address families by using the "show ospfv3 database" command.

```
R3#show ospfv3 database

           OSPFv3 1 address-family ipv4 (router-id 3.3.3.3)

                Router Link States (Area 1)

ADV Router       Age         Seq#        Fragment ID  Link count  Bits
 2.2.2.2         1312        0x80000005  0            1           B
 3.3.3.3         1262        0x80000005  0            1           None

                Net Link States (Area 1)

ADV Router       Age         Seq#        Link ID    Rtr count
 2.2.2.2         1312        0x80000001  4          2

                Inter Area Prefix Link States (Area 1)

ADV Router       Age         Seq#        Prefix
 2.2.2.2         1505        0x80000001  1.1.1.1/32
 2.2.2.2         1505        0x80000001  172.16.1.0/30
 2.2.2.2         1505        0x80000001  10.1.1.0/24

                Link (Type-8) Link States (Area 1)

ADV Router       Age         Seq#        Link ID    Interface
 3.3.3.3         1302        0x80000001  4          Gi0/2
 2.2.2.2         1505        0x80000001  4          Gi0/1
 3.3.3.3         1316        0x80000001  3          Gi0/1

                Intra Area Prefix Link States (Area 1)

ADV Router       Age         Seq#        Link ID    Ref-lstype  Ref-LSID
```

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```
 2.2.2.2         1312        0x80000003  0           0x2001      0
 2.2.2.2         1312        0x80000001  4096        0x2002      4
 3.3.3.3         1291        0x80000004  0           0x2001      0

          OSPFv3 1 address-family ipv6 (router-id 3.3.3.3)

                Router Link States (Area 1)

ADV Router       Age         Seq#        Fragment ID  Link count  Bits
 2.2.2.2         1310        0x80000004  0            1           B
 3.3.3.3         1259        0x80000004  0            1           None

                Net Link States (Area 1)

ADV Router       Age         Seq#        Link ID    Rtr count
 2.2.2.2         1310        0x80000001  4          2

                Inter Area Prefix Link States (Area 1)

ADV Router       Age         Seq#        Prefix
 2.2.2.2         1499        0x80000001  2005::1111/128
 2.2.2.2         1499        0x80000001  2002::/64
 2.2.2.2         1499        0x80000001  2001::/64

                Link (Type-8) Link States (Area 1)

ADV Router       Age         Seq#        Link ID    Interface
 3.3.3.3         1300        0x80000001  4          Gi0/2
 2.2.2.2         1499        0x80000001  4          Gi0/1
 3.3.3.3         1310        0x80000001  3          Gi0/1

                Intra Area Prefix Link States (Area 1)

ADV Router       Age         Seq#        Link ID    Ref-lstype  Ref-LSID
 2.2.2.2         1310        0x80000003  0          0x2001      0
 2.2.2.2         1310        0x80000001  4096       0x2002      4
 3.3.3.3         1288        0x80000004  0          0x2001      0
```

(##Notice that we have a database for each of our address families.)

Step 6: On R3, let's see if we can see our IPv4 routes by using the "show ip route" command. Let's also see if we are able to ping R1's IPv4 loopback interface.

```
R3#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
```

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```
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static
route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

      1.0.0.0/32 is subnetted, 1 subnets
O IA     1.1.1.1 [110/2] via 192.168.1.1, 00:01:08, GigabitEthernet0/1
      2.0.0.0/32 is subnetted, 1 subnets
O        2.2.2.2 [110/1] via 192.168.1.1, 00:01:08, GigabitEthernet0/1
      3.0.0.0/32 is subnetted, 1 subnets
C        3.3.3.3 is directly connected, Loopback0
      10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
O IA     10.1.1.0/24 [110/3] via 192.168.1.1, 00:01:08, GigabitEthernet0/1
C        10.2.2.0/24 is directly connected, GigabitEthernet0/2
L        10.2.2.1/32 is directly connected, GigabitEthernet0/2
      172.16.0.0/30 is subnetted, 1 subnets
O IA     172.16.1.0 [110/2] via 192.168.1.1, 00:01:08, GigabitEthernet0/1
      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.1.0/30 is directly connected, GigabitEthernet0/1
L        192.168.1.2/32 is directly connected, GigabitEthernet0/1
R3#ping 1.1.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 1.1.1.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 7/11/23 ms
```

(##Notice how we have knowledge of each of the IPv4 loopback addresses of each of our routers within R3's routing table.)
(##Notice how we were able to successfully ping R1's IPv4 loopback interface.)

Step 7: On R3, let's see if we can see our IPv6 routes by using the "show ipv6 route" command. Let's also see if we are able to ping R1's IPv6 loopback interface.

```
R3#show ipv6 route
IPv6 Routing Table - default - 11 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
       B - BGP, HA - Home Agent, MR - Mobile Router, R - RIP
       H - NHRP, I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea
       IS - ISIS summary, D - EIGRP, EX - EIGRP external, NM - NEMO
       ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
       RL - RPL, O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1
       OE2 - OSPF ext 2, ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
       la - LISP alt, lr - LISP site-registrations, ld - LISP dyn-eid
       lA - LISP away, a - Application
OI  2001::/64 [110/3]
     via FE80::5054:FF:FE14:6EE8, GigabitEthernet0/1
OI  2002::/64 [110/2]
     via FE80::5054:FF:FE14:6EE8, GigabitEthernet0/1
C   2003::/64 [0/0]
     via GigabitEthernet0/1, directly connected
```

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```
L   2003::2/128 [0/0]
     via GigabitEthernet0/1, receive
C   2004::/64 [0/0]
     via GigabitEthernet0/2, directly connected
L   2004::1/128 [0/0]
     via GigabitEthernet0/2, receive
OI  2005::1111/128 [110/2]
     via FE80::5054:FF:FE14:6EE8, GigabitEthernet0/1
O   2006::2222/128 [110/1]
     via FE80::5054:FF:FE14:6EE8, GigabitEthernet0/1
C   2007::/64 [0/0]
     via Loopback0, directly connected
L   2007::3333/128 [0/0]
     via Loopback0, receive
L   FF00::/8 [0/0]
     via Null0, receive
R3#ping 2005::1111
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2005::1111, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 5/15/41 ms
```

(##Notice how we have knowledge of each of the IPv6 loopback addresses of each of our routers within R3's routing table.)
(##Notice how we were able to successfully ping R1's IPv6 loopback interface.)

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